

Math709 – Foundations of Computational Mathematics II

Spring 2013

Homework 1

1. Let $\mathbf{A} \in \mathbb{C}^{m \times m}$ be hermitian, i.e., $\mathbf{A}^* = \mathbf{A}$.
 - (a) Prove that all eigenvalues of $\mathbf{A}$ are real.
 - (b) Prove that if $\mathbf{x}$ and $\mathbf{y}$ are eigenvectors corresponding to distinct eigenvalues, then $\mathbf{x}$ and $\mathbf{y}$ are orthogonal.
2. Let $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{A} \in \mathbb{C}^{m \times n}$, show that
 - (a) $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \sqrt{m}\|\mathbf{x}\|_\infty$.
 - (b) $\frac{1}{\sqrt{n}}\|\mathbf{A}\|_\infty \leq \|\mathbf{A}\|_2 \leq \sqrt{m}\|\mathbf{A}\|_\infty$.

3. Determine SVDs of the following matrices by hand calculation:

$$(a) \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} \quad (b) \begin{bmatrix} 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (c) \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

4. Given $\mathbf{A} \in \mathbb{C}^{m \times n}$ with $m \geq n$, show that $\mathbf{A}^*\mathbf{A}$ is nonsingular if and only if $\mathbf{A}$ has full rank.
- 5.* Suppose $\mathbf{A} \in \mathbb{C}^{m \times m}$ has an SVD $\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^*$. Find an eigenvalue decomposition of the $2m \times 2m$ hermitian matrix

$$\begin{bmatrix} \mathbf{0} & \mathbf{A}^* \\ \mathbf{A} & \mathbf{0} \end{bmatrix}$$

6. Consider the matrices

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- (a) What is the orthogonal projector $\mathbf{P}$ onto $\text{range}(\mathbf{A})$, and what is the image under $\mathbf{P}$ of the vector $(1, 2, 3)^*$?
 - (b) Same questions for $\mathbf{B}$.
7. Let $\mathbf{P} \in \mathbb{C}^{m \times m}$ be a nonzero projector. Show that $\|\mathbf{P}\|_2 \geq 1$, with equality if and only if $\mathbf{P}$ is an orthogonal projector.